

CSC 476 – PROJECT 1 (SPACE ROCKS)

Technical Report

Chaughn Robin



How To Play

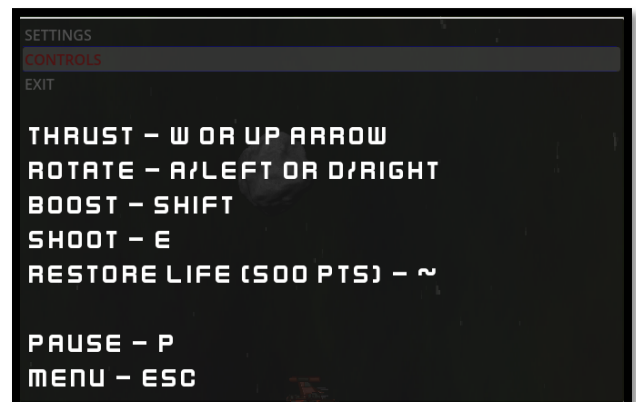
Menu –

On launch, the game begins on the main menu screen. The options are ‘Play’, ‘Settings’, and ‘Exit’.

- Press **Play** to begin the game.
- Press **Settings** to open the settings menu.
- Press **Exit** to exit the game.

Controls –

- **‘WASD keys or ↑←↓→ keys’ are used for the movement controls.**
- **W / ↑** moves the player forward in the facing direction.
- **S / ↓** moves the player opposite the facing direction.
- **D / →** rotates the player to the right.
- **A / ←** rotates the player to the left.
- **ESC** opens or closes the menu overlay. If the game is in progress, opening the menu will also pause the game.
- **P** pauses the game.
- **E** fires the ship’s lasers.
- **Shift** engages the ship’s boosters.



- The controls can also be found by navigating to **Main Menu→Settings→Controls** or by using the menu overlay's **Controls** tab.



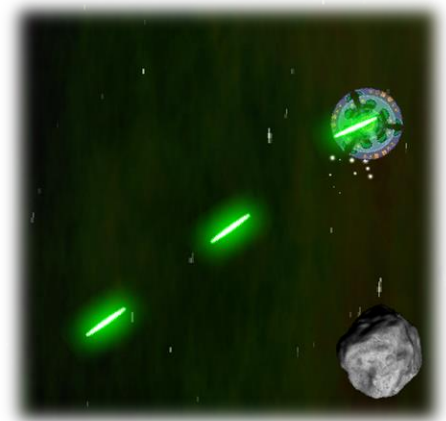
Obstacles –

Asteroids are one of the examples of space debris you can encounter as you explore the vastness of deep space.



Enemies –

Hostile Ships are one of the most common enemies when traversing space. They will fire in bursts at the player, and commonly take **3 shots** to destroy. They will collide with each other, obstacles, and the player. **Hostile Ships** can also be destroyed by each other's friendly fire.



Results of the Playtesting

Phase2:

- Menu functionality has been added. Has settings, control, and exit buttons/items.
- Controls item in menu stays selected, requiring the player to select another item, then controls again to make it pop back up.

- Several laser upgrades are available.

- Powerups have been added. Laser powerups and health and shield powerups.

- Laser powerups:
 - Hyperfire : Dramatic fire rate increase
 - Death laser: Fire a solid beam of laser death!
 - Damage upgrade: Turns regular lasers into supercharged versions. 1 shot kill, additional penetration! Laser body disappears after going through multiple bodies.
 - Spread Fire: Fires a laser with an increased spread. *Not fully functional*

- Enemies and players both show explosions when shot, adding to the effects.

- Added additional sounds for enemies lasers, and several laser power up sounds.

- Did not get to boss fights.

Phase1:

- Navigation between starting screen to game start, and game end back to starting screen is functioning correctly.
- More enemies and obstacles needed to add more depth to game.
- Currently, no boss fights, but boss fights will be either every 5 or 10 levels.
- Total # of powerups are limited and powerups are spawned by chance upon destruction of asteroids.
- Added shield bar and functionality. Shield depletes before health. If shield == 0, health will deplete.

Documentation

Four changes from the Project instructions:

1. Change the theme of your game by changing the look and sounds of your game.
2. Add an additional dynamic element of the game that rewards the player somehow. (Powerups with unique looks, sound, and animations. Black holes,)
3. Add an additional dynamic element of the game that harms the player somehow.
4. Creating a second game experience for a single level after a set number of levels or other triggering event.

5. Create a story between levels with animation, sound, and text.

6. Create a secondary player that is inert while not being used, but that can be activated by the player for use rendering the primary player inert.

❖ - Yellow highlight indicates this has not yet been implemented.